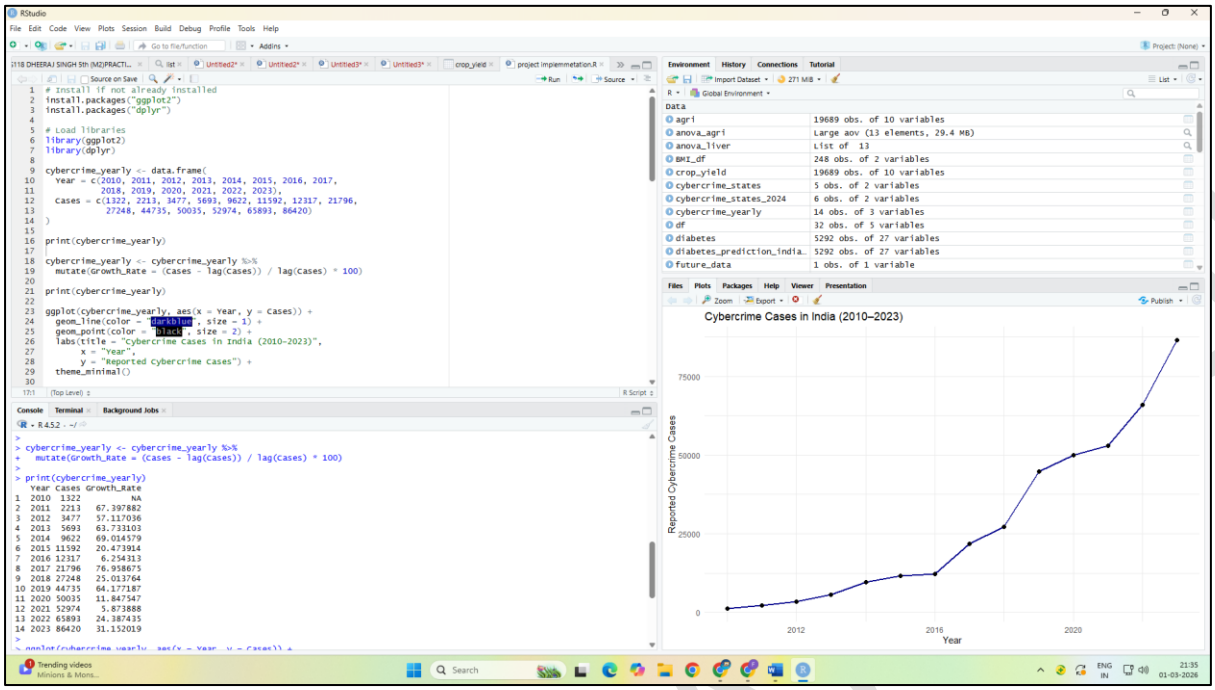
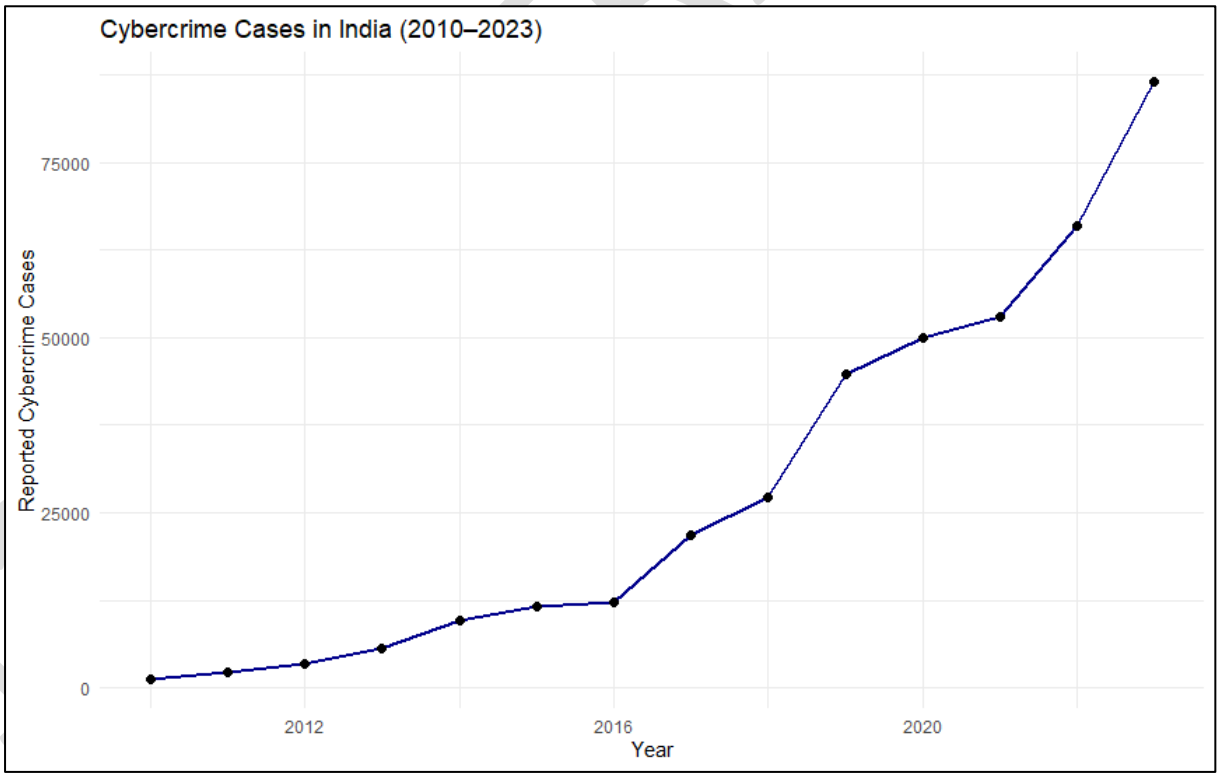
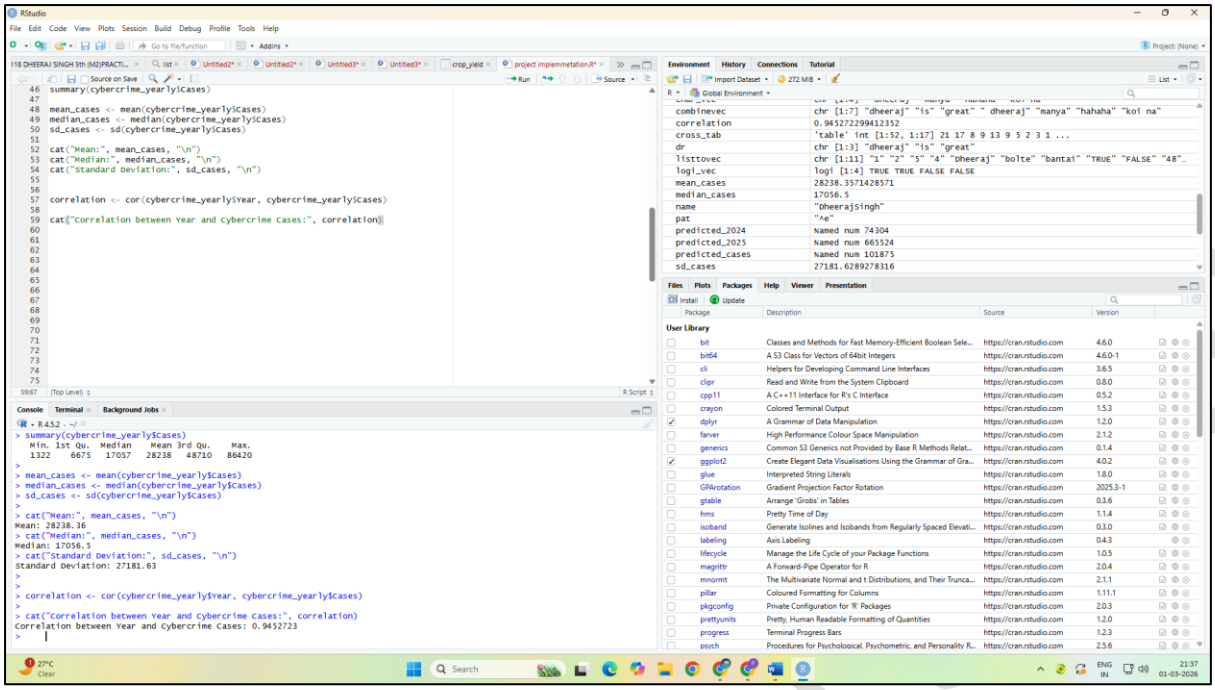
Implementation of Research using R







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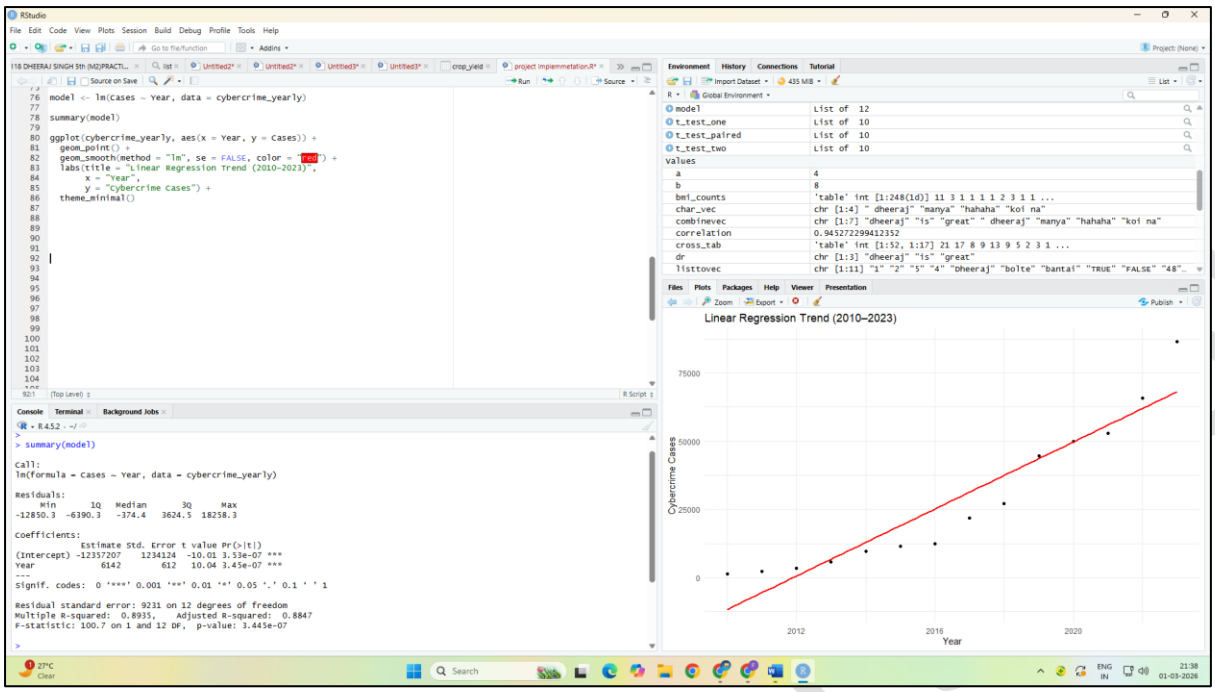
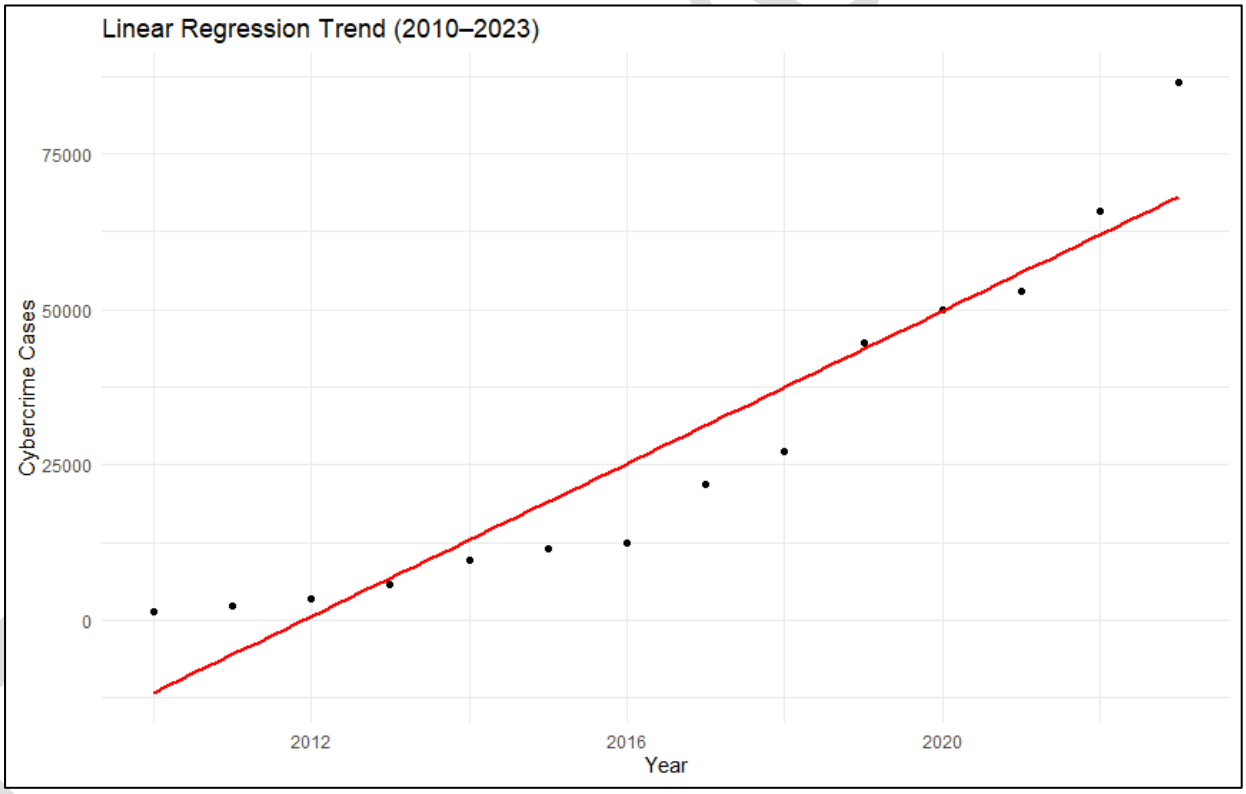
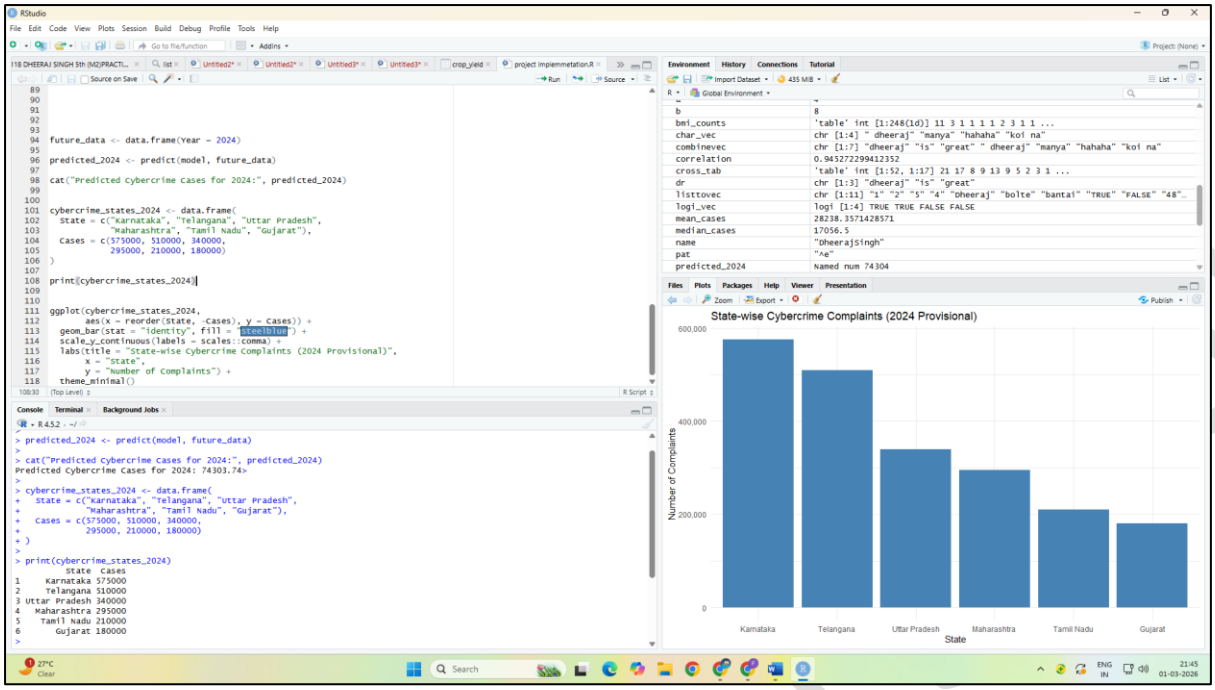
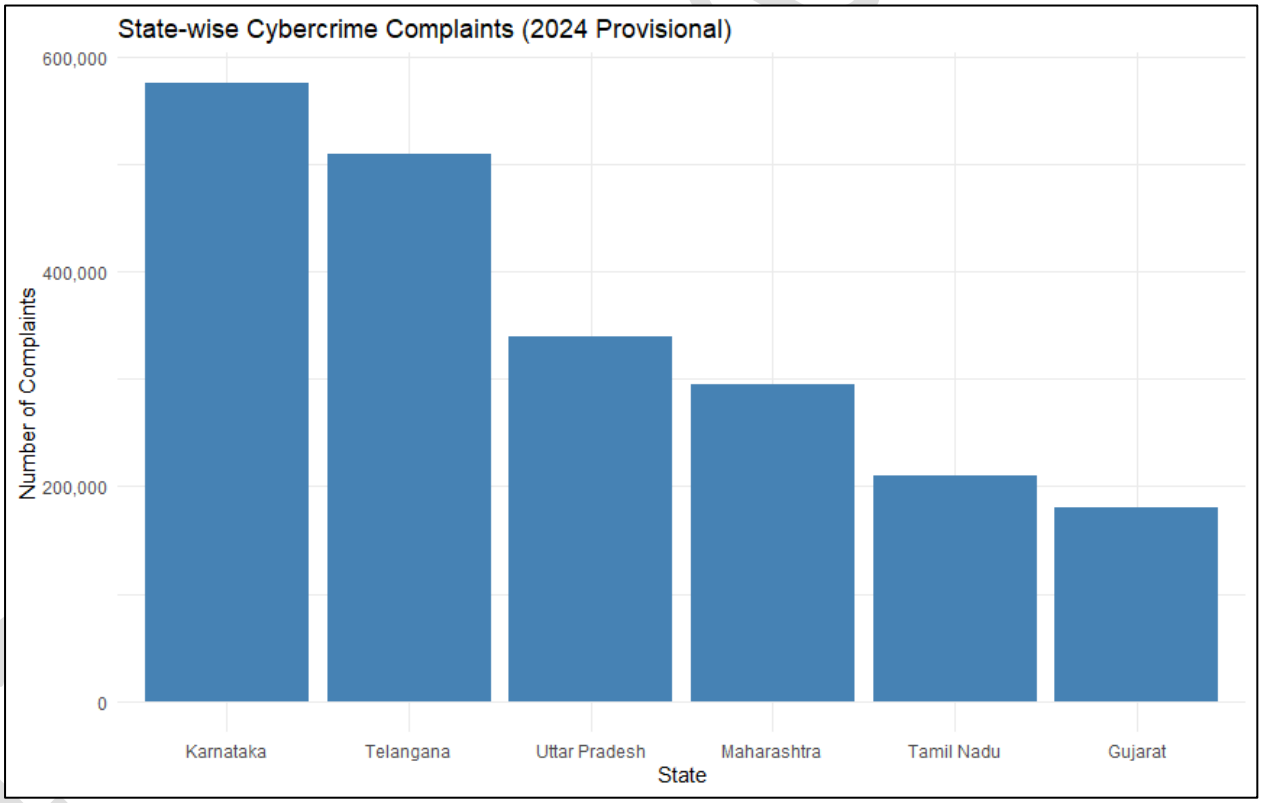

model <- lm(Cases ~ Year, data = cybercrime_yearly)
summary(model)
ggplot(cybercrime_yearly, aes(x = Year, y = Cases)) +
geom_point() +
geom_smooth(method = "lm", se = FALSE, color = "red") +
labs(title = "Linear Regression Trend (2010–2023)",
x = "Year",
y = "Cybercrime Cases") +
theme_minimal()
> summary(model)
Call:
lm(formula = Cases ~ Year, data = cybercrime_yearly)
Residuals:
Min 1Q Median 3Q Max
-12850.3 -6390.3 -374.4 3624.5 18258.3
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) -12357207 1234124 -10.01 3.53e-07 ***
Year 6142 612 10.04 3.45e-07 ***
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Residual standard error: 9231 on 12 degrees of freedom
Multiple R-squared: 0.8935, Adjusted R-squared: 0.8847
F-statistic: 100.7 on 1 and 12 DF, p-value: 3.445e-07
Linear Regression Trend (2010–2023)
Cybercrime Cases
Year


Linear Regression Trend (2010–2023)
75000
50000
25000
0
Cybercrime Cases
2012
2016
2020
Year


RStudio
future_data <- data.frame(year = 2024)
predicted_2024 <- predict(model, future_data)
cat("Predicted Cybercrime Cases for 2024:", predicted_2024)
cybercrime_states_2024 <- data.frame(
State = c("Karnataka", "Telangana", "Uttar Pradesh",
"Maharashtra", "Tamil Nadu", "Gujarat"),
Cases = c(575000, 510000, 340000,
295000, 210000, 180000)
)
print(cybercrime_states_2024)
ggplot(cybercrime_states_2024,
aes(x = reorder(State, -Cases), y = Cases)) +
geom_bar(stat = "identity", fill = steelblue) +
scale_y_continuous(labels = scales::comma) +
labs(title = "State-wise Cybercrime Complaints (2024 Provisional)",
x = "State",
y = "Number of Complaints") +
theme_minimal()
> predicted_2024 <- predict(model, future_data)
> cat("Predicted Cybercrime Cases for 2024:", predicted_2024)
Predicted Cybercrime Cases for 2024: 74303.74
> cybercrime_states_2024 <- data.frame(
+ State = c("Karnataka", "Telangana", "Uttar Pradesh",
+ "Maharashtra", "Tamil Nadu", "Gujarat"),
+ Cases = c(575000, 510000, 340000,
+ 295000, 210000, 180000)
+)
> print(cybercrime_states_2024)
State Cases
1 Karnataka 575000
2 Telangana 510000
3 Uttar Pradesh 340000
4 Maharashtra 295000
5 Tamil Nadu 210000
6 Gujarat 180000
State-wise Cybercrime Complaints (2024 Provisional)
600,000
400,000
200,000
0
Number of Complaints
Karnataka
Telangana
Uttar Pradesh
Maharashtra
Tamil Nadu
Gujarat
State


State-wise Cybercrime Complaints (2024 Provisional)
600,000
400,000
200,000
0
Number of Complaints
Karnataka
Telangana
Uttar Pradesh
Maharashtra
Tamil Nadu
Gujarat
State

CODE Link:

https://github.com/dheeraj-coder48/RESEARCH-USING-R

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